Futurist Robot User Manual

1. Safety

1.1 Safety Notices

- Before using the robot, read the user manual carefully to understand the robot's functions, operating methods, and safety precautions.
- Use the robot on a level surface. Avoid extremely high or low temperatures, humid environments, steep slopes, large drops, fragile surfaces, and areas with vibration.
- Ensure that the robot has enough operating space. Avoid narrow or crowded environments to reduce the risk of collision and pinching.
- All robots are equipped with an emergency stop device. Users must be familiar with the location and use of the emergency stop during operation so that the robot can be stopped promptly in an emergency. Please note that after emergency stop is triggered, the robot joints will collapse into a limp state. Do not use it unless there is a real emergency or unless the robot is adequately protected.
- Before operation, make sure all cables, plugs, and sockets are intact. In case of abnormal conditions such as short circuit or overheating, disconnect the power immediately.
- Perform regular maintenance and inspections to ensure that components such as joints, sensors, and power systems are operating normally, and to avoid hazards caused by aging or damage.
- Prevent misuse. Ensure that the robot is not used in any improper, dangerous, or legally prohibited scenarios.

1.2 Safety Guidelines

- Before startup, confirm the following:
  - Before starting the robot, make sure there are no people or other obstacles in the working area, especially near moving parts.
  - Before startup, the operator should confirm that the system status is normal and self-check has been completed.
- Maintain a safe distance from the robot's working range to avoid accidental collisions, especially while the robot is performing high-speed motions.
- Payload and operation:

  - Strictly follow the robot's design load limits and avoid overload operation.
  - When using the robot's actuators, grippers, or other components, ensure they are matched to the target object to avoid unnecessary pressure and stress.
- If an accident occurs or the robot loses control, first use the stop function on the controller. If the controller fails, press the emergency stop button, then inspect the equipment and environment after power-off.
- During robot operation, do not directly interfere with its movements. Any adjustment or manual intervention must be performed only after the robot has completely stopped and power has been disconnected.
- Avoid incorrect operation. Ensure that all operators receive relevant training and understand how to operate the robot correctly and how to handle emergencies.
- Remote operation safety:
  - If you develop or use remote operation features, ensure a secure network environment to prevent loss of control caused by network failure or external intrusion.
  - During remote control, the operator must have real-time monitoring equipment and full visibility of the robot and its surroundings.

1.3 Maintenance and Management Guidelines

- Perform regular maintenance according to FF-EAI Robotics' recommendations to keep the robot in optimal working condition.
- Fault handling:
  - If the robot malfunctions during operation, stop operation immediately and notify qualified technical personnel for inspection and repair.
  - Do not disassemble or repair the robot while it is running.
- Battery and power management:
  - If the robot uses a battery, ensure that the battery is charged in a safe and dry environment.
  - Do not leave the robot connected to power for an extended time to avoid overcharging or overheating.
- Keep the robot operating system and control software updated regularly to prevent vulnerabilities and safety risks.

Note: Following this manual helps ensure safe and efficient robot operation. If you have any questions, please contact the manufacturer or an authorized service provider for support.

WARNING

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

Note:

This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions:

(1) This device may not cause harmful interference, and

(2) this device must accept any interference received, including interference that may cause undesired operation.

2. Basic Description

2.1 Product Model

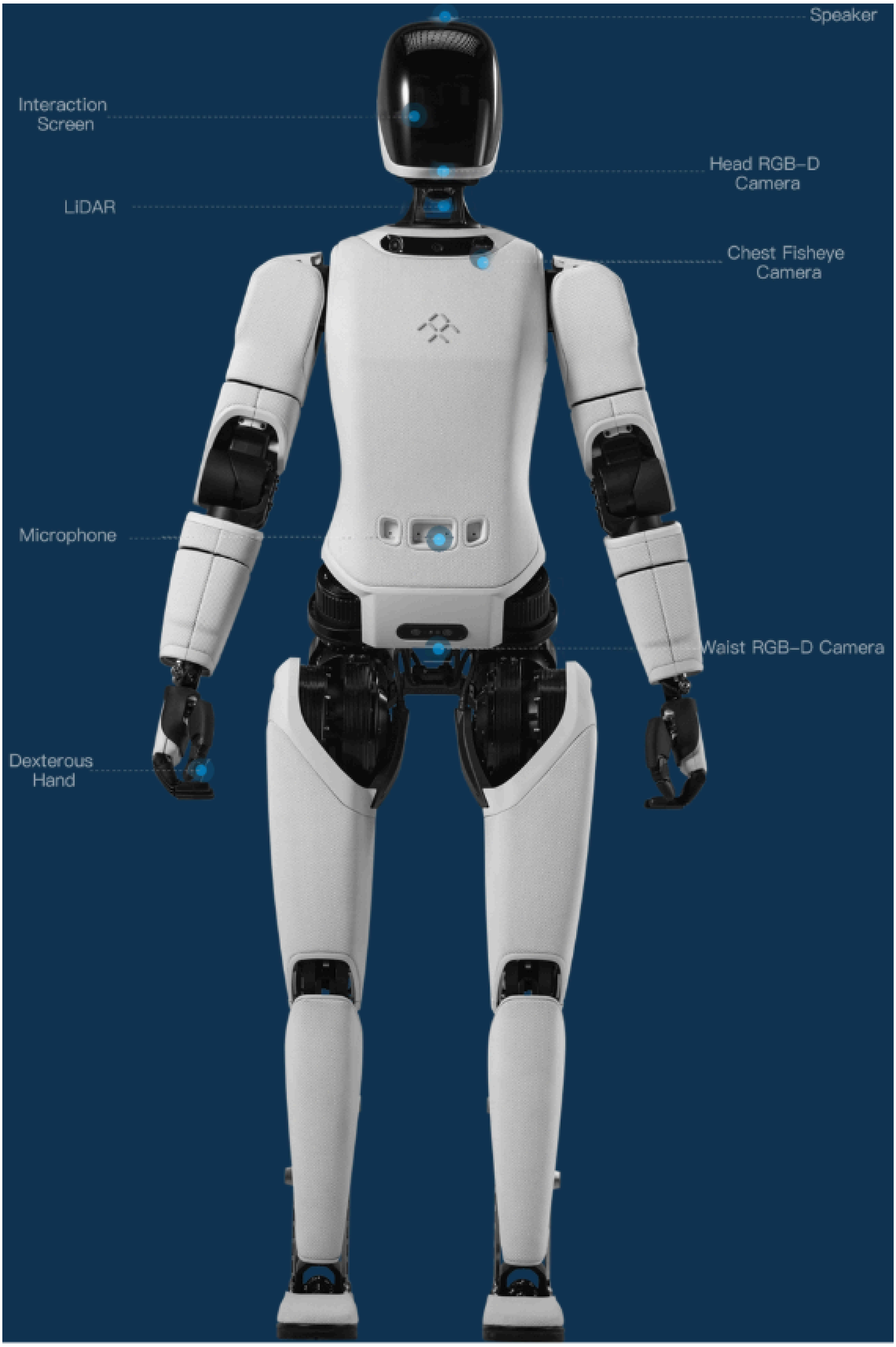
1. **Naming rule:**

Brand name	FF-EAI Robotics
Model	Futurist

2. Model lookup: After the device is connected successfully, click [About] in the AimMaster settings interface to view the current product model.

2.2 Product Composition

The source manual includes a front and rear structure diagram identifying the major robot components, including the speaker, arm joints J1-J5, soft prosthetic hand, head rotation joint, shoulder ring, battery, maintenance compartment, leg joints J1-J6, and depth camera.



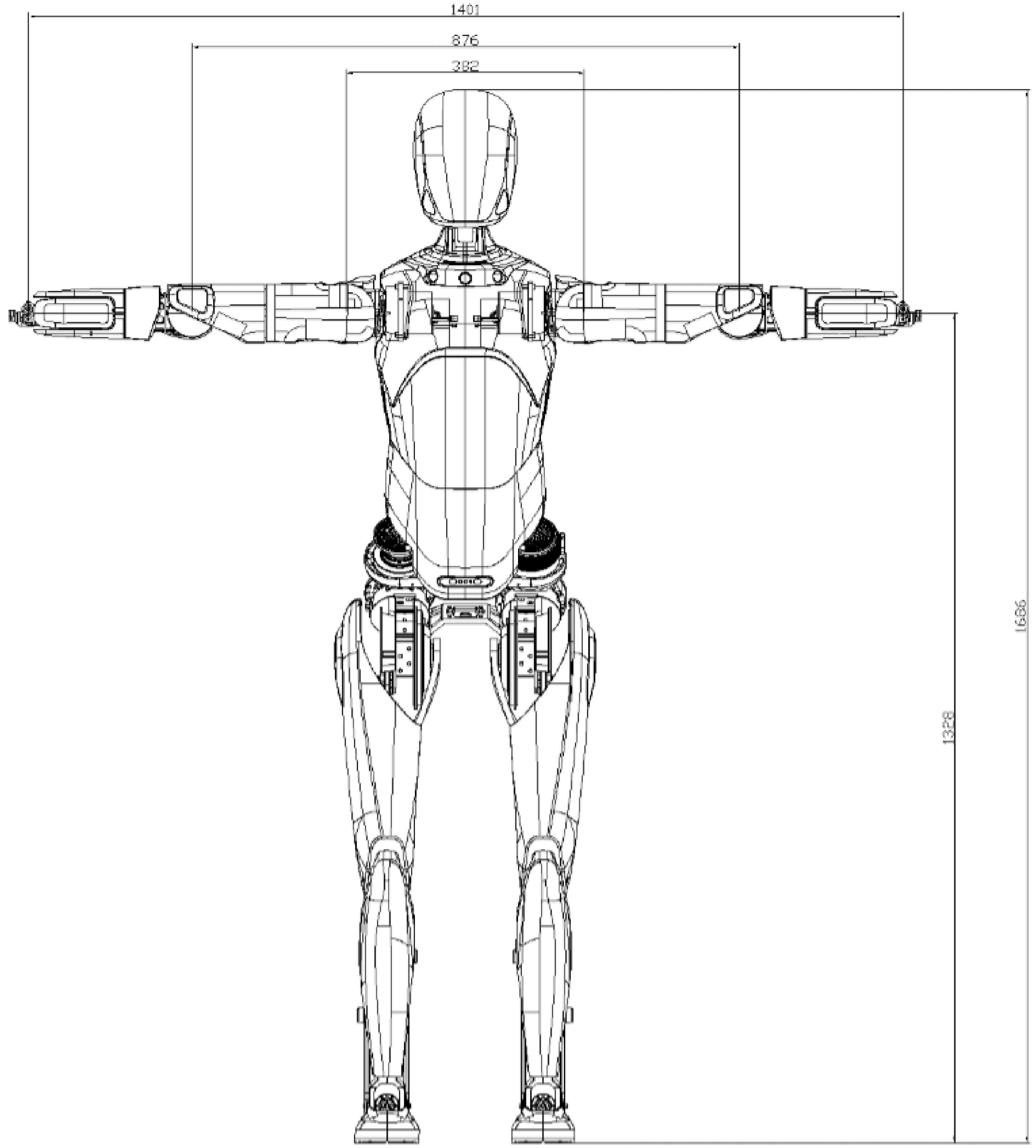
2.3 Basic Specifications

Level 1 Category	Level 2 Category	Item	Details
Product basic information		Height	169 cm
		Dimensions	169(H) x 75(W) x 30(L) cm
		Net weight	Approx. 63 kg
		Actuated DoF	Dual arms: 5 x 2 DoF; dual legs: 6 x 2 DoF; head: 1 D
Product characteristics	Electrical performance	Endurance	Force-controlled standing endurance: 4 h 35 min; endurance: 1 h 35 min
		Battery capacity	14.4 Ah
		Charging time	2 h
		Charging power	<= 500 W
		Charging input voltage	110 V - 220 V
		Charger output	54.6 V 8 A
	Environmental adaptability	Operating temperature / humidity	0 - 40 C, relative humidity 10% - 90%, non-conden
		Storage temperature / humidity	-20 - 70 C, relative humidity 10% - 90%, non-conde
		IP rating	Joint modules: IP5X
	Terrain adaptability	Minimum passable width	Current minimum width under body remote contr minimum width under navigation > 2 m
		Maximum obstacle-crossing height	20 mm
		Maximum operating slope	8% (4.57 degrees)
	Operation and interaction	Perception capability	Not supported
		Voice capability	Not supported
		Light-duty operation capability	Not supported
		Remote operation	Wireless remote controller
		Speaker	5 W
		Dance development	Supported
		Group control software	Supported
	Mobility	Maximum movement speed	Maximum speed 0.8 m/s; <= 0.6 m/s in daily use
		Movement modes	Supports translation, diagonal movement, and in- turning
	IoT capability	Communication protocol	TCP/IP
		Communication module	Wi-Fi
	Intelligent control parameters	Basic computing board	16-core high-performance CPU
	Typical arm parameters	Dexterous hand	Soft prosthetic hand
		Rated payload per arm	1.5 kg (excluding end effector)
		End-effector linear speed	1 m/s

		Arm motion range	J1 (Shoulder pitch): ±170°; J2 (Shoulder roll): -30° (Shoulder yaw): ±170°; J4 (Elbow pitch): -1° to 118 (Wrist roll): ±170°
	Typical leg parameters	Leg motion range (with slight deviations)	J1 (Hip roll): -37° to 40°; J2 (Hip yaw): ±75°; J3 (Hi -50° to 110°; J4 (Knee pitch): -5° to 120°; J5 (Ankle to 52°; J6 (Ankle roll): ±28°
	Typical head parameters	Head motion range	Rotary joint: ±45°

2.4 Overall Working Envelope

The source manual provides overall front, top, and side working-envelope drawings of the robot. These are reproduced in the appendix for visual reference.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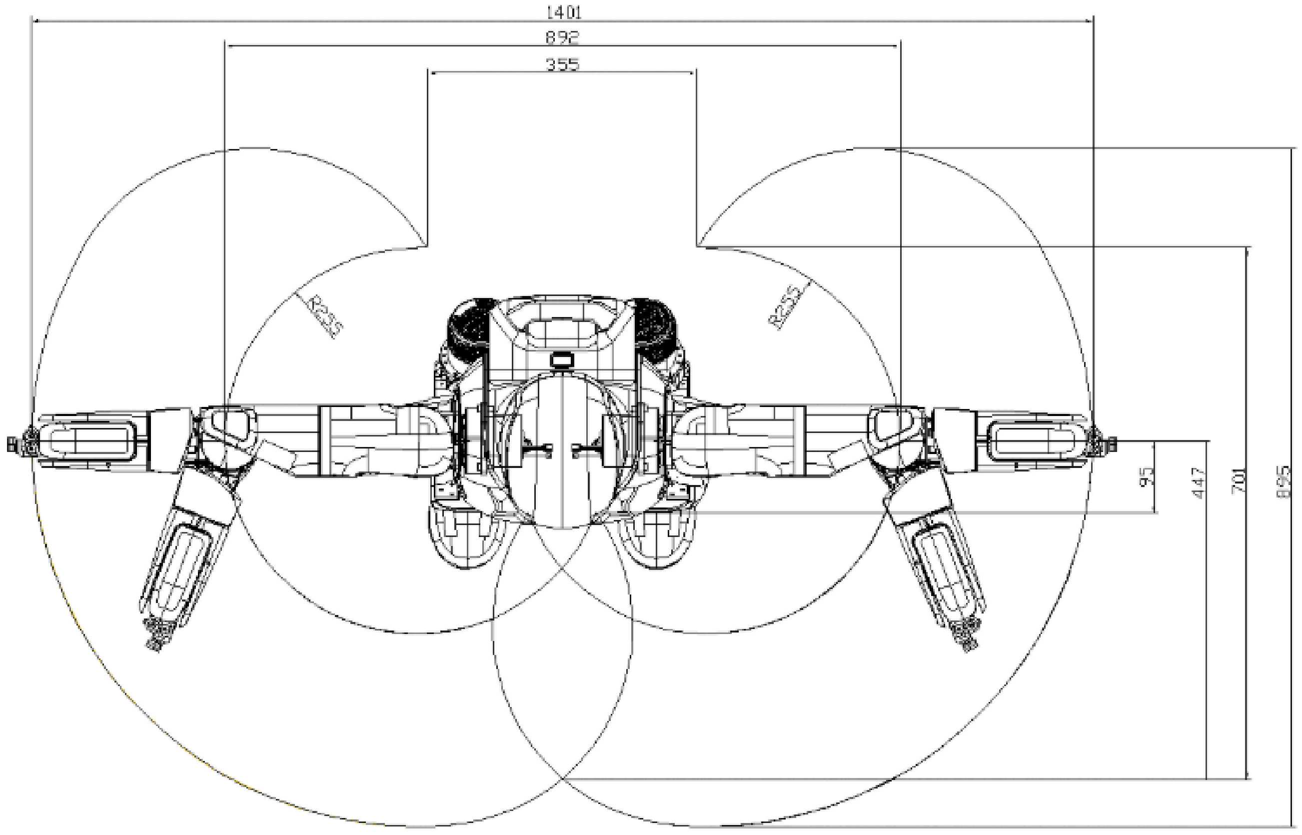
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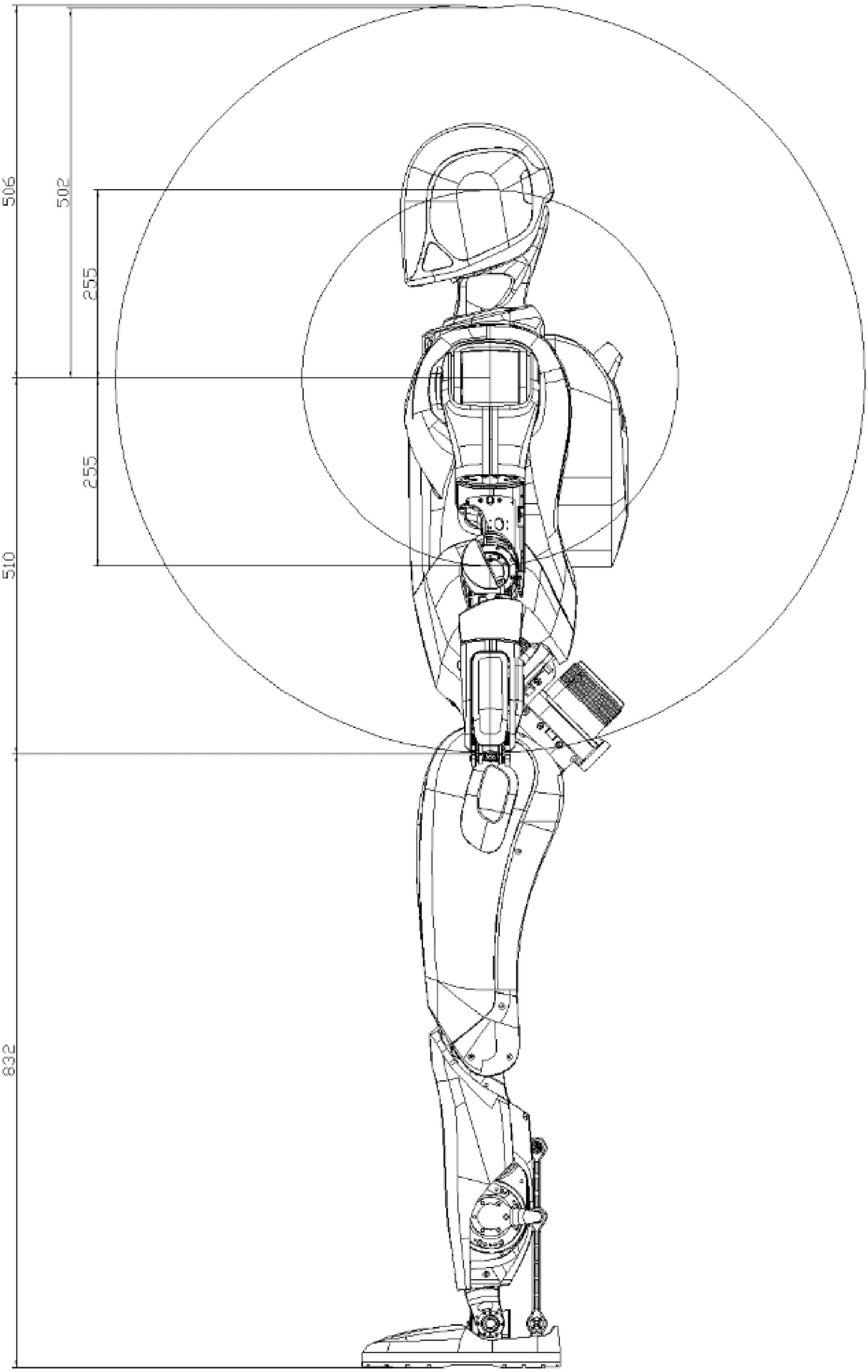


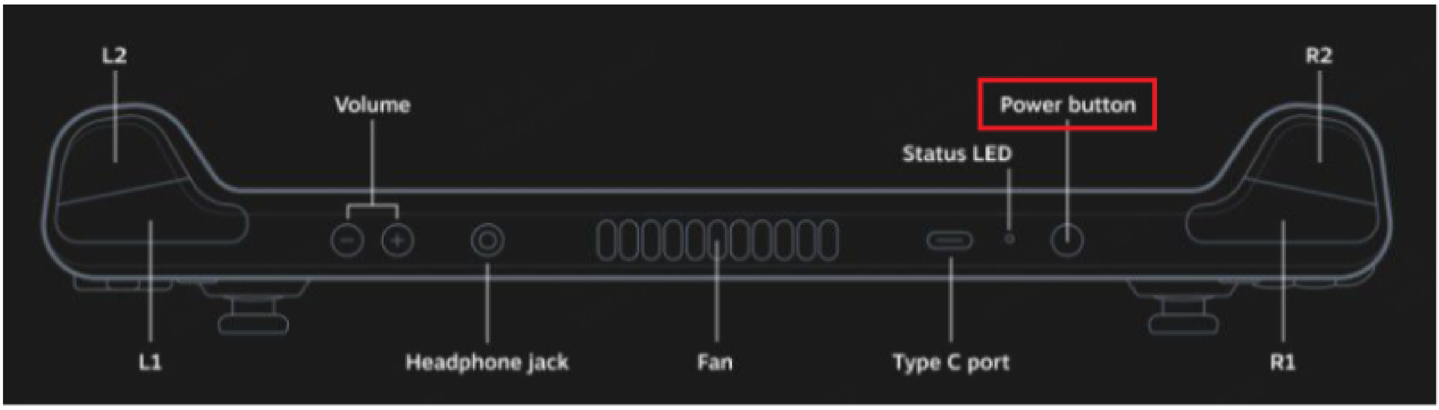
3. Quick Start Guide

Before performing the following operations, first complete robot unboxing according to the unboxing document.

I. Power On

Step 1: Open the AimMaster Client

Turn on the official FF-EAI Robotics controller. The controller power button is located at the top. Press and hold the power button until the white light next to the power indicator turns on. After the controller starts up, AimMaster opens automatically and enters the robot startup wizard. The controller charging port is located at the bottom of the remote.

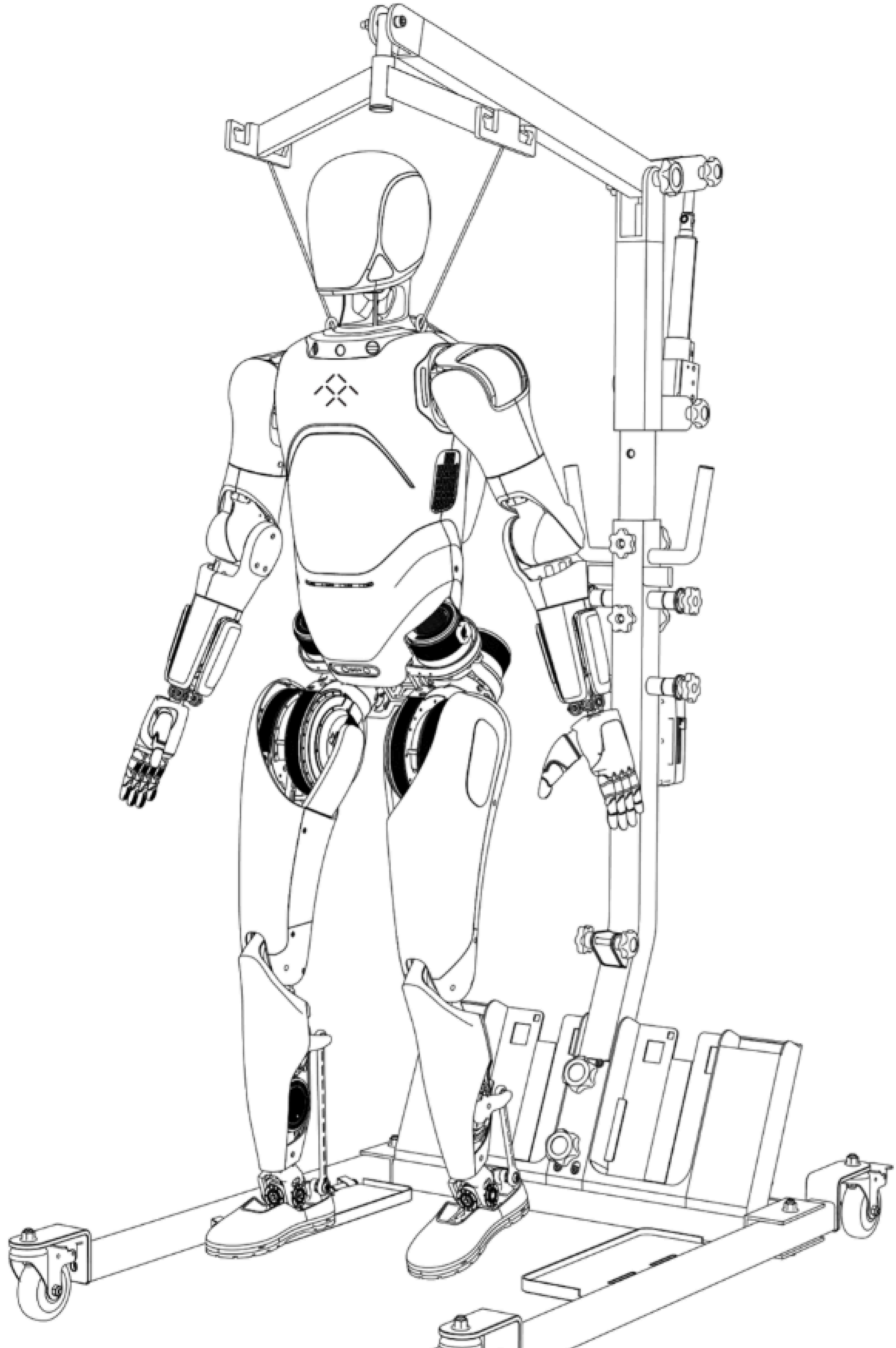


On the robot type selection screen, choose “FF Futurist”, then enter the FF Futurist startup wizard and follow the prompts step by step to complete power-on.

Step 2: Hoist the Robot

Before powering on the robot, the robot must first be hoisted. Connect the robot to the hook at the top of the electric lift using the shoulder lifting ring and sling, then manually operate the electric lift until the robot is in the position shown and both feet are completely off the ground.

Note: While moving or hoisting the robot, take care not to squeeze or strike the dexterous hands.



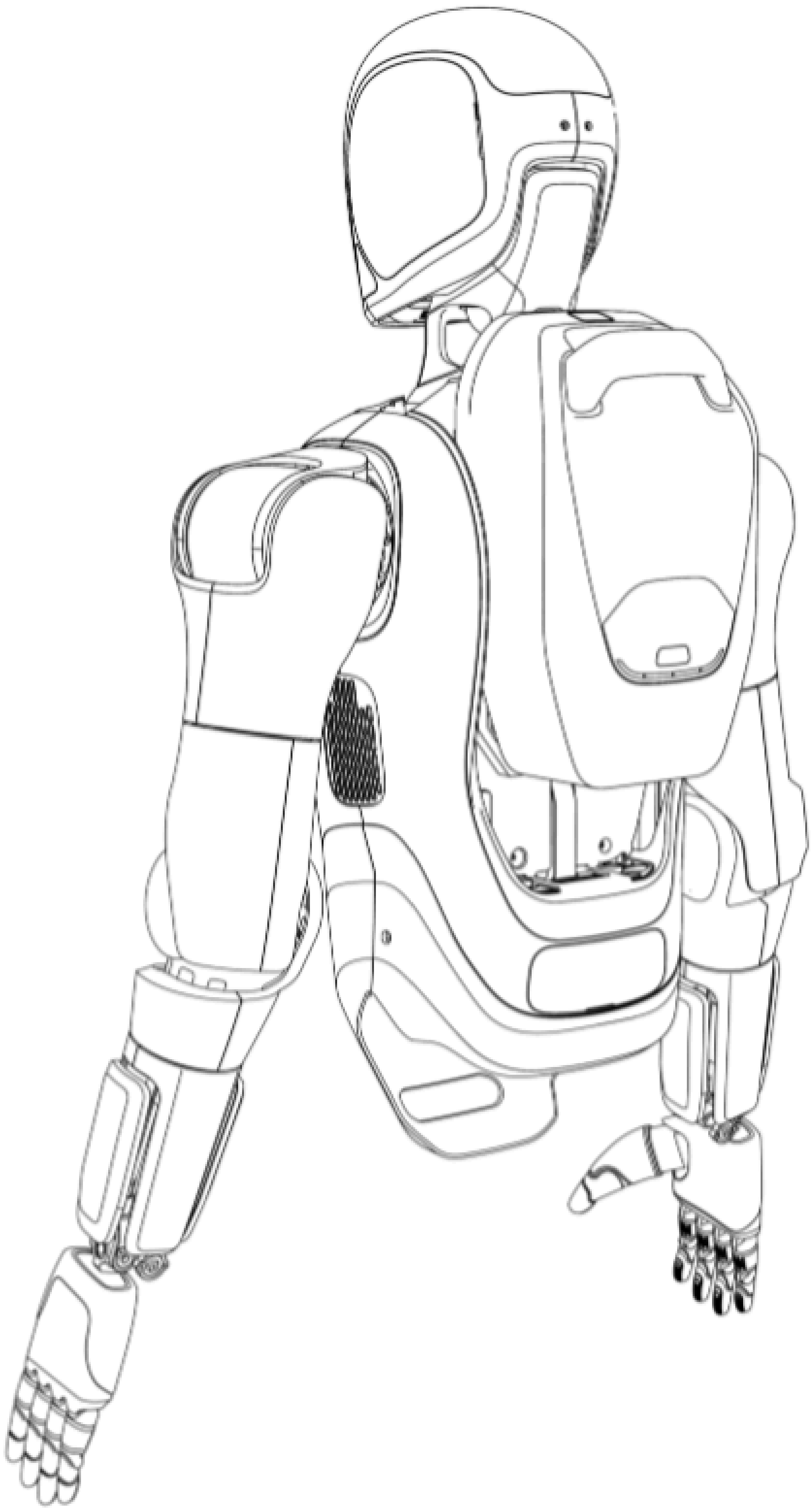
After the robot is hoisted, click “Next” in the AimMaster interface.

Step 3: Install the Battery and Power On

The battery is installed on the back of the robot. Confirm that the battery has sufficient charge, then slide it down from top to bottom into the battery slot on the robot's back. When the battery reaches the bottom, you will hear a “click,” indicating that it is fully locked in place.

Press and hold the power button in the maintenance compartment at the rear side of the robot's waist to power on the robot. When power is connected, the LED inside the power button lights up.

Power button operation: press and hold to power on; to power off, short-press once and then immediately press and hold.



After startup is complete, click “Next” in AimMaster.

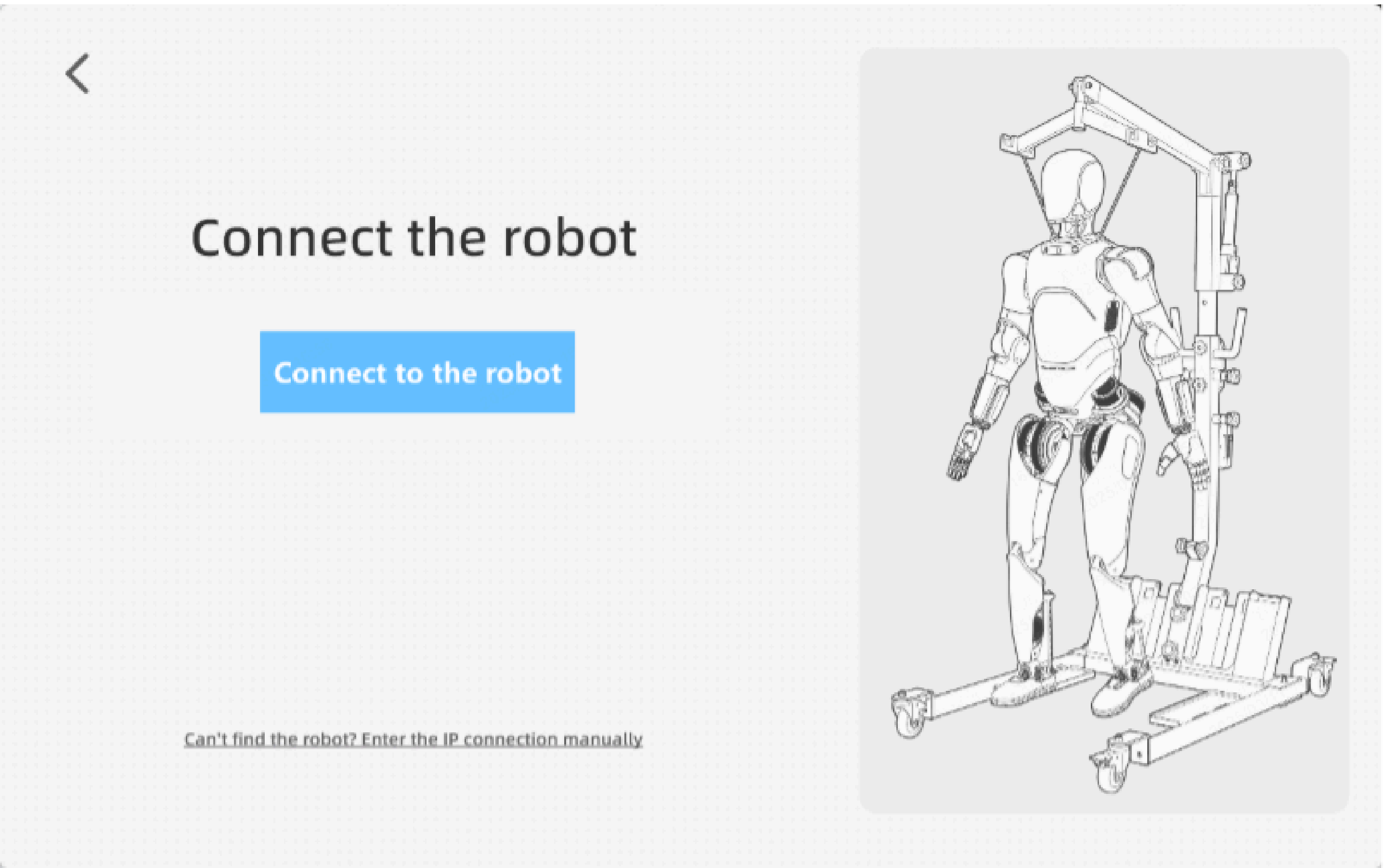
Step 4: Wireless Emergency Stop

The emergency-stop remote is simple to use and is paired before shipment, so it works out of the box. In the default released state, the motors are powered normally. In an emergency, press the button to cut power to the motors. After the issue has been resolved, rotate the button to release it and re-energize the motors. A green indicator means normal operation; red indicates an abnormal state. When fully charged, the remote displays four battery indicator LEDs.

During the robot joint self-check, the neck will move slightly for self-inspection. After the joint self-check is complete, click “Next” in AimMaster.

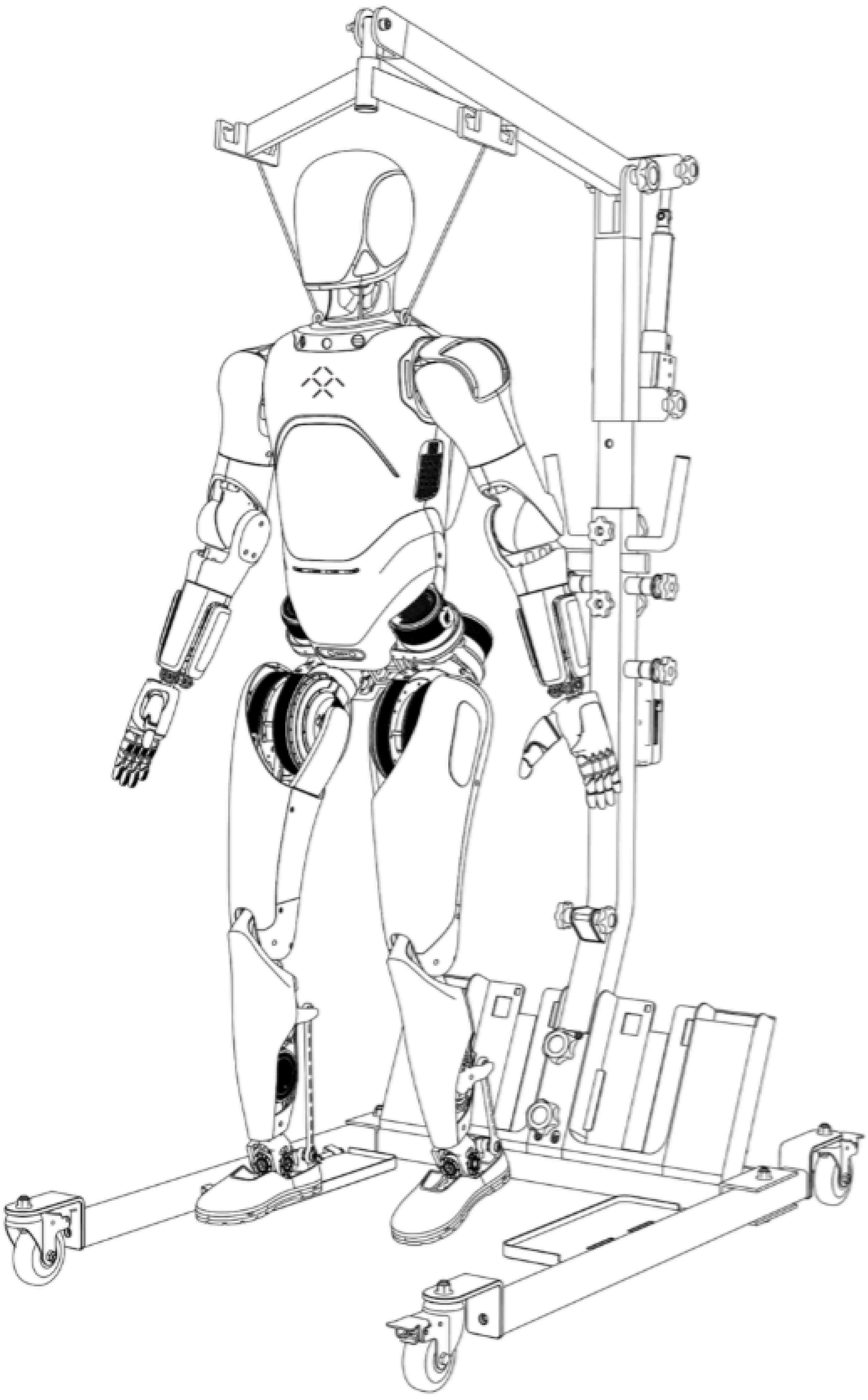
Step 5: Connect the Robot

After the emergency stop has been released, AimMaster shows the robot connection screen. Click the connection button to connect or manually enter the IP address to connect.



Connection method and safety verification:

- Recommended before first use: For safety, after using the remote for the first time, set a robot authorization password first to prevent unauthorized devices from connecting.
- Robot IP connection:
- Click the prompt text at the bottom of the page to open the input box, then manually enter the robot IP address.
- A password input box appears. Enter the correct password to connect.
- If the password is incorrect, the system displays: "Connection exception, please re-enter password."
- A password must be entered for each IP connection; the password is stored on the device.
- The robot IP can be viewed under [Settings] - [Wireless LAN].



- View / change password:
  - Go to [Settings] - [Authorization Password] to view or modify the password.
  - Enter a new password (at least 8 characters), then click “Confirm.”

After the robot is successfully connected, AimMaster displays “Connection successful.” Click “Next.”

Step 6: Switch to the Standing Preparation Pose

Before the robot stands up, first switch it to the standing preparation pose while it is still suspended. In the “Switch Standing Pose” screen in AimMaster, click the “Switch Pose” button. The robot automatically adjusts the positions of the head, arms, and legs to the preparation pose.

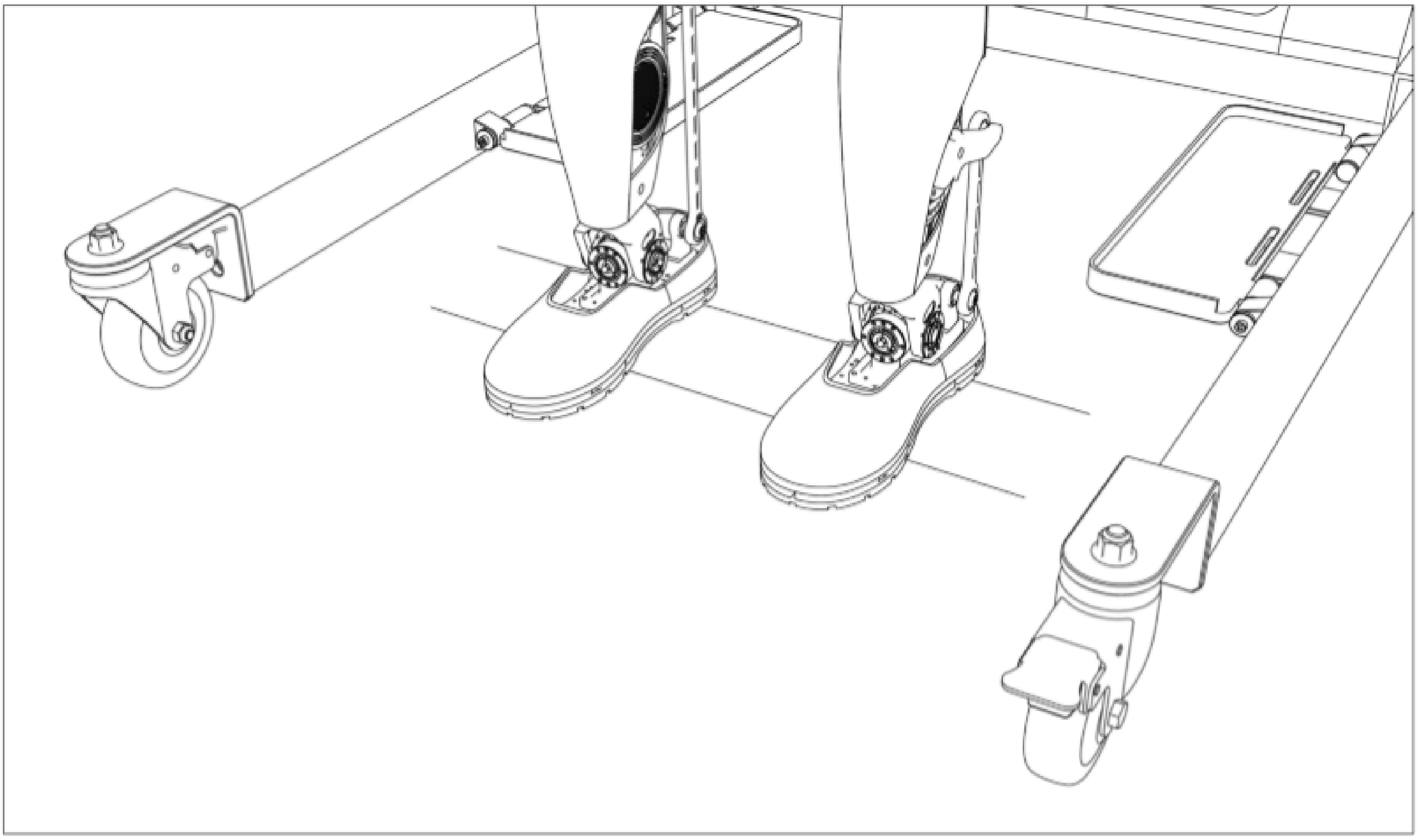
In the preparation pose, both arms are slightly bent, the wrists are parallel to the forearms, both legs are slightly bent, and the soles are approximately parallel to the floor.

Note: The joints will move during the pose-switching process. Stay away from the robot and make sure there is sufficient space around it.

After the standing preparation pose has been completed, click “Next.”

Step 7: Lower the Robot

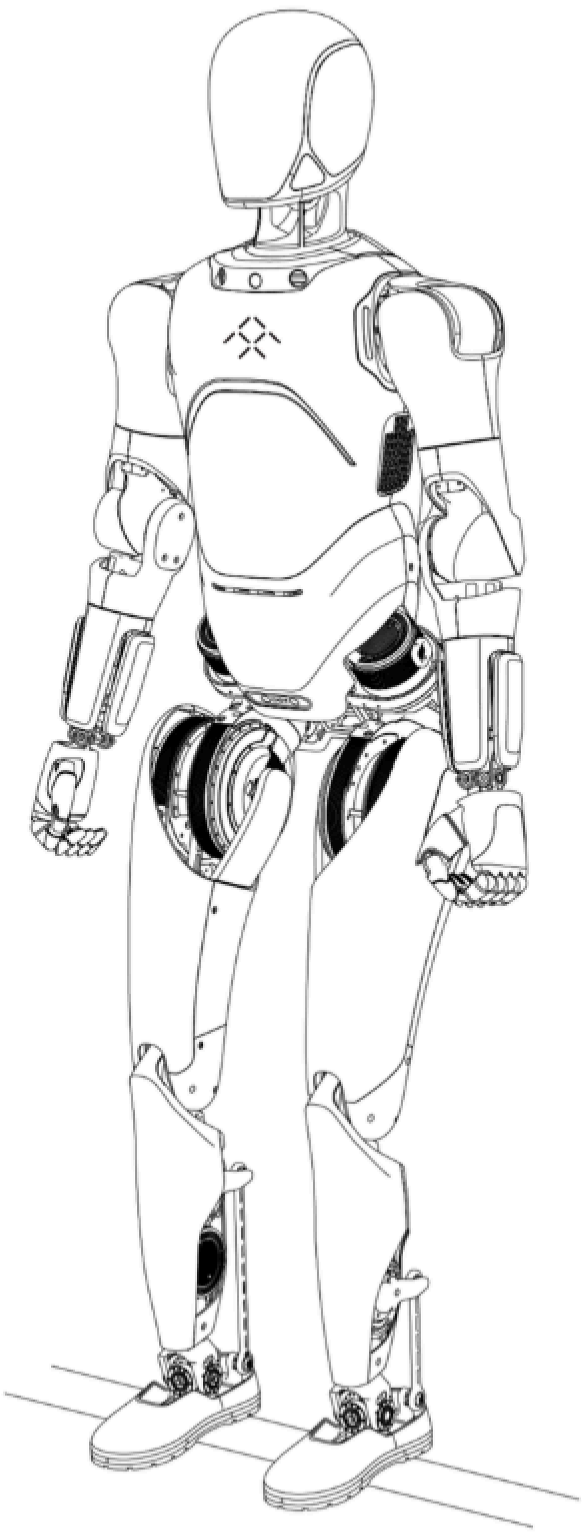
Instructions for the electric lift: Place the electric lift behind the robot and ensure that the robot stands between the two legs of the lift. After confirming safety, press the lower button on the lift and observe the lowering process. Continue until both feet are fully in contact with the ground and the suspension sling is completely slack with a little extra slack.



After the lowering operation is complete, click “Next” in AimMaster.

Step 8: Switch to Standing Mode

On the “Switch Standing Mode” screen in AimMaster, click the “Enable Standing Mode” button. The robot enters standing mode. When the message “Standing Successful!” appears, the standing process is complete. At this point, the shoulder sling may be removed, and the robot will maintain autonomous balanced standing.



Note: Before switching to standing mode, make sure the robot has already been lowered to the ground and both feet are fully contacting the floor.

Click "Finish" to exit the legged startup wizard; AimMaster will enter the workbench page.

II. Power Off

Step 1: Install the Lifting Ring

Raise the lifting arm of the standby frame until the robot's feet are completely off the ground.

Step 2: Hoist the Robot

Install the lifting rings at the robot's shoulders and use the electric lift to hoist the robot until both feet are completely off the ground.

Step 3: Power Off

Press the power button once inside the maintenance compartment at the rear side of the robot's waist to cut power and shut down the robot.

III. Motion Control

3.1 Walking Preparation

Before using the remote to walk the FF Futurist robot, first ensure that the robot has completed the standing procedure (refer to the standing process in the power-on section). After the robot is standing, click the Toolbox panel on the left side of the AimMaster main screen to view the mode-switching function bar.

Double-click to switch to "Walking, Natural Arm Swing" mode. Following the flow shown in the interface, click to execute the relevant actions and the robot will begin stepping in place. Switch to "Motion" and the robot can perform interaction actions.

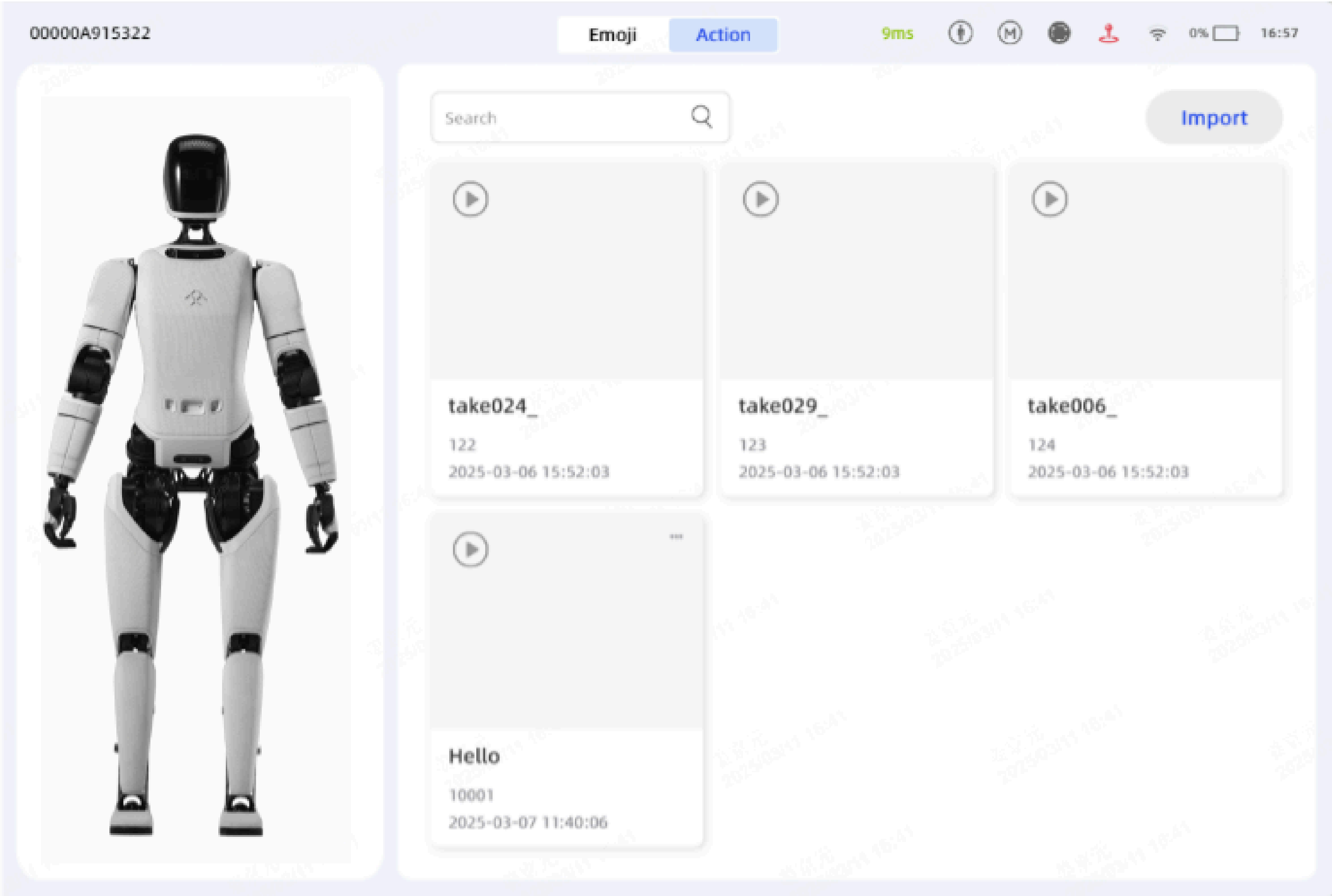
3.2 Remote Walking

Click the "Control" button on the software sidebar to enter the "Remote Walking" page.

3.3 Interactive Actions

Click "Interaction" in the sidebar to switch to the interaction page. Click "Action List" at the top of the screen to enter the action list page. Swipe up and down to view the preset robot actions stored on the robot. Click the play button in the upper-left corner of a card to play the selected action, and the robot will begin performing it.

Click "Custom Broadcast" at the top of the screen to enter the custom broadcast list page, which allows users to create new broadcast content. Swipe to browse the broadcast list; double-click an item to play it.



4. Charging and Battery Replacement

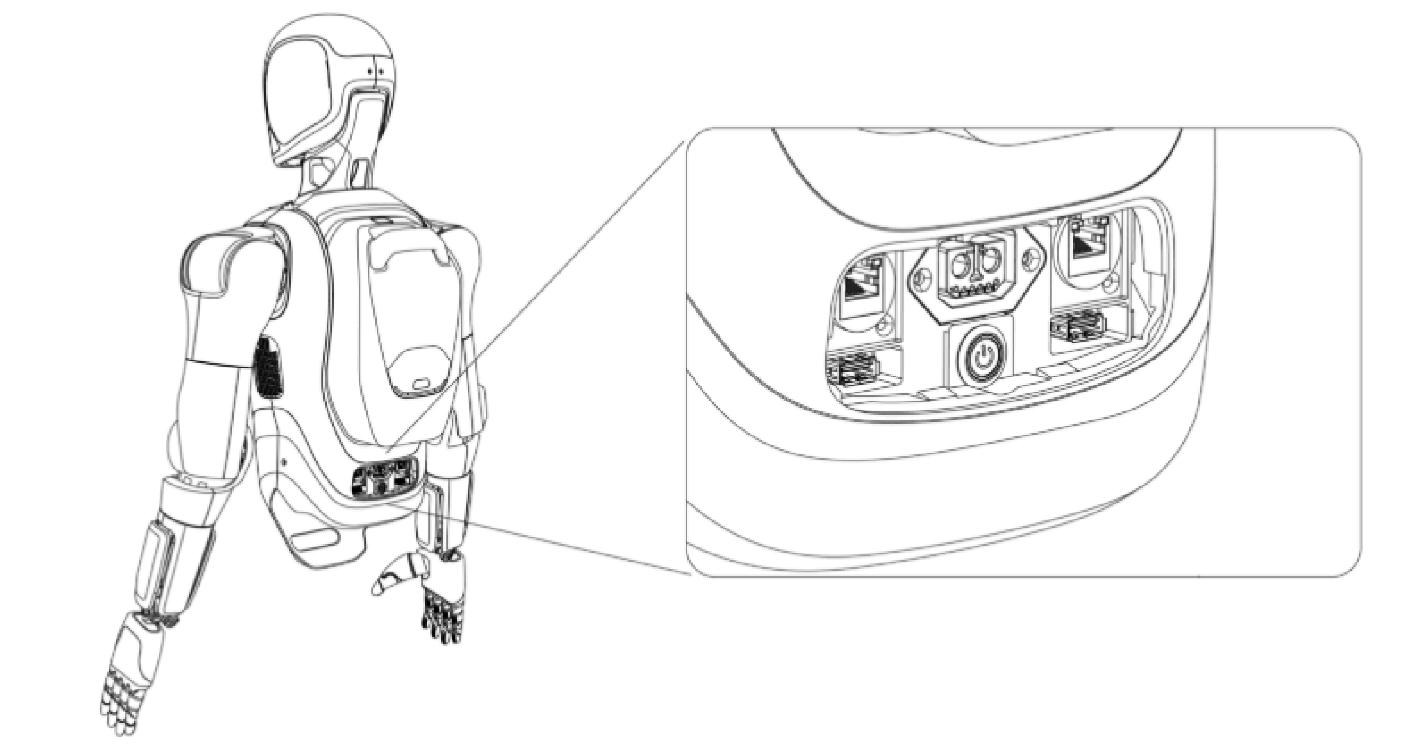
4.1 Charging

Step 1: Open the Maintenance Compartment

Open the maintenance compartment cover on the back of the robot.

Step 2: Insert the Charging Connector

The charging port is located below the rear-side cover at the waist of the robot. First remove the cover, then insert the charger plug from the power adapter into the charging port.



4.2 Battery Replacement

Step 1: Power Off

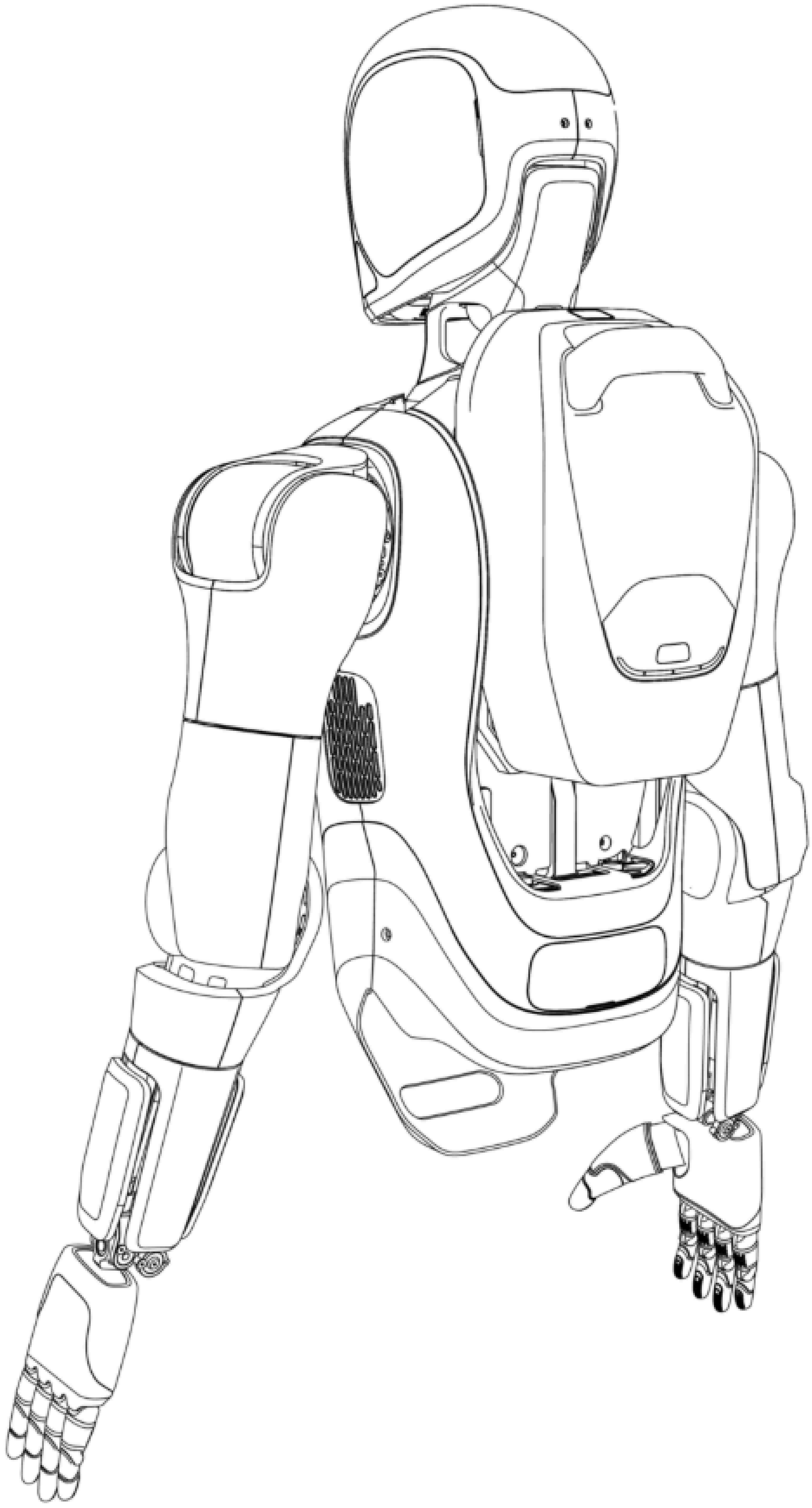
Refer to the power-off procedure in Section 3.

Step 2: Remove the Battery

Press and hold the square mechanical release button on the top of the battery with your thumb while lifting the battery upward with your four fingers.

Step 3: Install the Battery

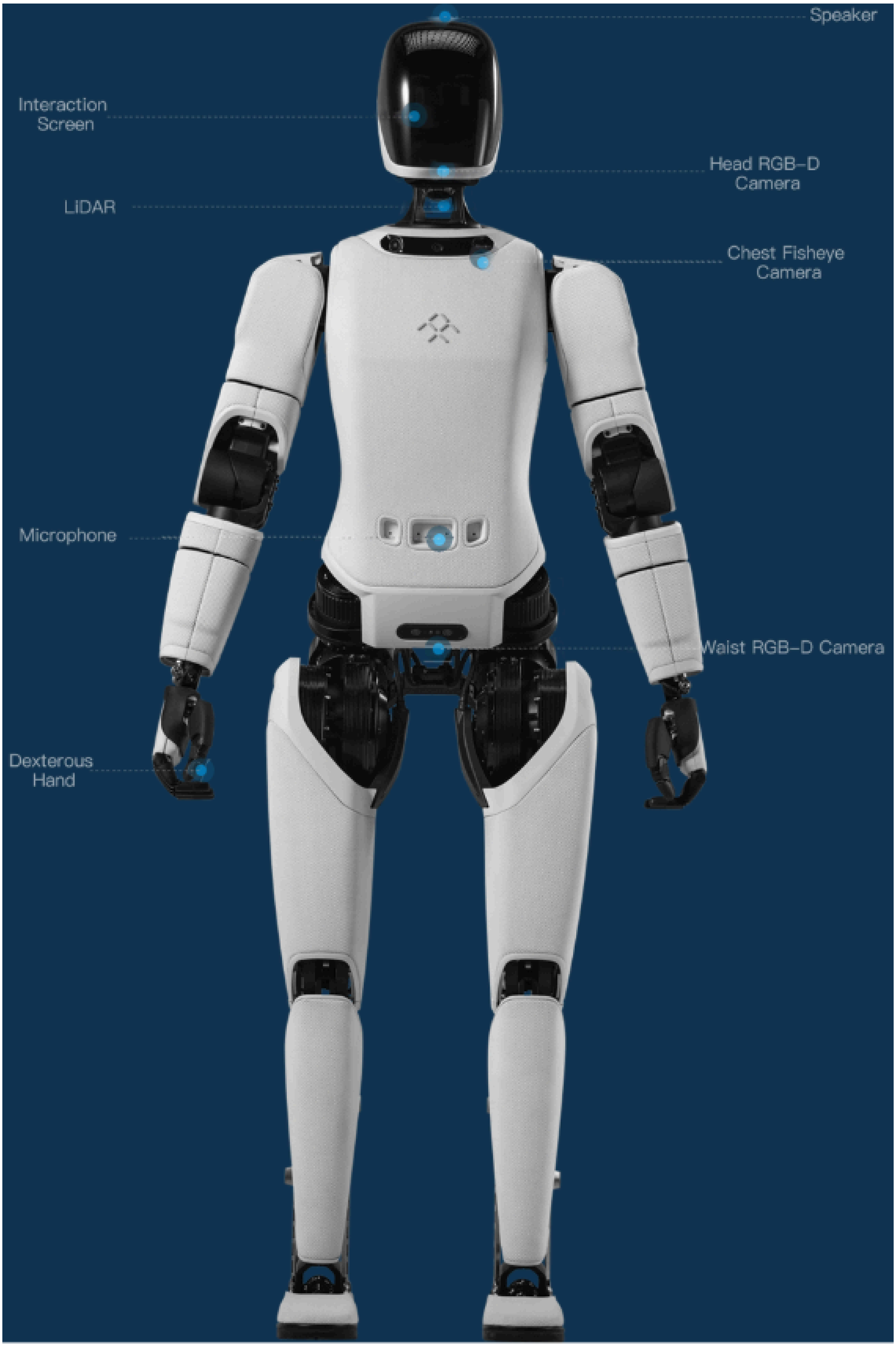
Refer to Step 3 under the power-on procedure to install the battery.



5. Dressing Guide

Core requirements for clothing: joints must not be wrapped, the fabric must not restrict movement, putting on and taking off should be easy to align, and heat dissipation must not be blocked.

- Reduce the impact on heat dissipation: Leg joints J1, J2, J5, and J6 should not be covered.
  - Loose shorts are recommended. At J1 and J2, local cutouts in the clothing are recommended to improve airflow.
- Weight limit: The total weight of clothing should be controlled within 2 kg.
  - Avoid heavy fabrics such as wool or thick fleece. Overly heavy materials directly increase the load on the limbs.
- Reduce restrictions on joint range of motion: Arm joints J3 and J4 should not be covered, and the linkage between the legs and feet should not be covered.
  - Loose, highly elastic clothing is recommended; avoid fabrics such as denim.
  - The shoe upper should not cover moving mechanisms, and the sole must provide good slip resistance.
- Ease of dressing and undressing: Avoid slim or tight-fitting cuts.
  - Prefer front-opening designs, such as front-opening tops, front-button jackets, wide necklines, and sleeveless tops.
  - Choose smooth-surfaced fabrics that are not prone to snagging.



6. Other Notes

- This manual is based on the Futurist Edition complete machine and software version v1.0.

- For any items not specified in this user manual, please visit the FF-EAI Robotics official website or contact FF-EAI Robotics service personnel for more information.